INCH-POUND

MIL-S-50673A (AR)
12 December 1990
SUPERSEDING
MIL-S-50673 (MU)
15 April 1971

MILITARY SPECIFICATION

SHAPER, PULSE: 10548806

This specification is approved for use within the U. S. Army Armament, Munitions and chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pulse shaper used in Radar Receiver-Transmitter RT-860, part of Radar Set AN/VPS-2, and in Subassembly Test Set MX-8048, part of Radar Test Set AN/TPM-22 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection of, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-81O - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assembles

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publicatications. The following other Government documents. drawings. and publications form a-part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548806 - Shaper, Pulse

PACKAGING DATA SHEET

, PS10548O6 - Packaging of Pulse Shaper

(Copies of drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548806, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the pulse shaper shall meet the requirements of MIL-P-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The pulse shaper shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be-3.75 minutes ± 100 per decade
The 10-55-10 hertz sweep time is 3 minutes 45 seconds ± 10%.

3.3.2 Shock. The pulse shaper shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The pulse shaper shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The pulse shaper shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

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3.3.4 Temperature. The pulse shaper shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 degrees F and +155 degrees F.

3.3.4.2 Non-operating (storage). The pulse shaper shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The pulse shaper shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the pulse shaper.

3.5 Performance. When supplied with the inputs and terminations specified in Table I, the pulse shaper shall yield the outputs specified in Table I (Input and output characteristics).

TABLE I. Input and output characteristics.

	REQUIREMENT	CHARACTERISTICS	TERMINALS
1.	Input power	29 ± 1 vdc	E6 (+) and E3 (ret)
2.	Input signal	PRF : 20.0 ± 0.1 KHz Pulse width: 0.3 ± 0.05 us Pulse amplitude: +10.0 ± 1.0 V into 50 ohm load.	El and E3 (ret)
3*	Terminating loads	50 ± 1 ohms each, to ground	E5, E4, E2
4*	Output signal	PRF : 20.0 +_0.1 K H z Pulse width; Not less than 0.5 us. Pulse amplitude: Not less than +15.0 v.	E5, E4, E2

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

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3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task)F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS .

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchases order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification **where such** inspections are deemed necessary to assure supplies and services conform to prescribed requirements .

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5 and all applicable drawings. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements **in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance** does not authorize submission of **known defective material, either indicated or actual, nor does it commit the** Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-P-11926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. **First article inspection (see 4.3).**
b. **Quality conformance inspection (see 4.4).**
c. **Special sampling inspection (see 4.5).**

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg.F to +90 deg.F
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

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4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) Shaper, Pulse for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any *or* all of the examinations and tests specified in Table 11 (Functional tests) and Table III (Special sampling) herein and to any or all requirements 'of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior *or* subsquent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

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4.4.2 Examination and tests.

4.4.2.1 Components, assemblies and subassemblies. all components, assemblies, and subassemblies shall be inspected in accordance with the inspection provisions contained in-this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be Performed on a single defect (individual characteristic) basis in accordance with-the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be Performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.

TABLE II. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One pulse shaper out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table 11 and shall then meet the requirements and tests of Table III. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

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TABLE 111. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit

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and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor Inspection equipment designs shall be submitted for **approval in accordance with 6.3.**

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IV (Test equipment list) and Table V (Test harness parts list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE IV. Test equipment list.

ITEM NUMBER	DESCRIPTION
1. Power supply	28 ± 1 Vdc; less than 50 mV peak-to-peak ripple; 500 mA capacity.
2. Pulse generator	Capable of delivering pulses 0.3 us wide, less than 12 ns rise time, +10 amplitude, at PRF of 20.0 ± 0.1 KHz, into a 50 ohm load.
3* Test harness	As specified in Figure 1. See Table V for test harness parts list.
4. oscilloscope	vertical passbend: DC to 24 MHz. Rise time: 15 ns. Input sensitivity: 50 mV/cm. Horizontal deflection: 0.2 us/cm to 5 s/cm.

TABLE V. Test harness parts list.

ITEM	NOMENCLATURE	DESCRIPTION
R1	Resistor	50 ohm, 5%, 1/2 w
R2	Resistor	50 ohm, 5%, 1/2 w
R3	Resistor	50 ohm, 5%, 1/2 w

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

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4.7.1.1 Vibration. **The test shall be** performed as specified in Method 514, Category 8, Procedure I, **of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only of paragraph II-2.1 apply. The pulse shaper shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the pulse shaper shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.**

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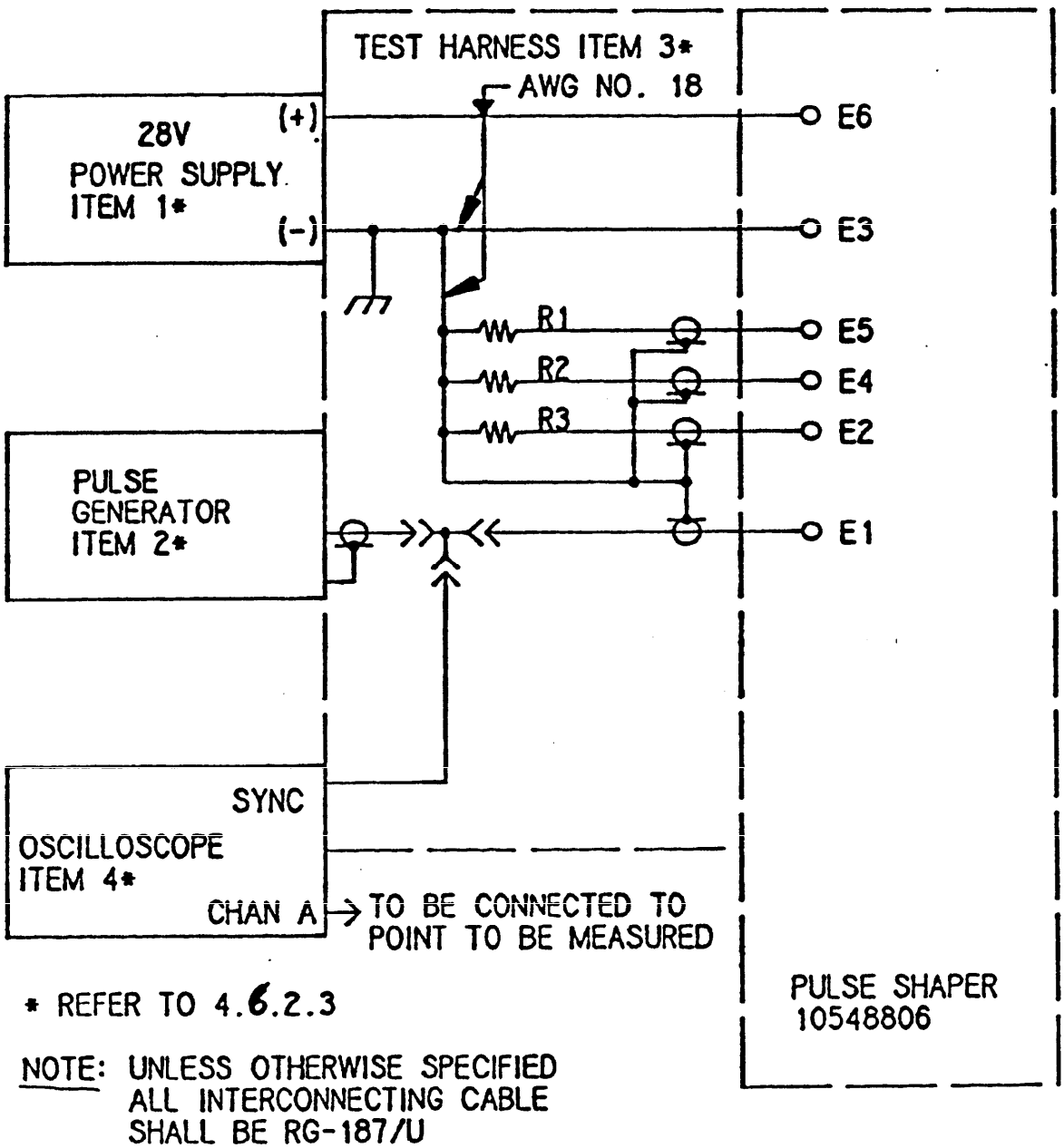


FIGURE 1. Pulse shaper test setup.

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4.7.1.2 Shock. This test shall be performed as specified in procedure I, Selection II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 Hertz.

After the application of all the shock impulses, the pulse shaper shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating.) This test shall be Performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified In 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be *performed as* specified in Procedures I and II"Methad 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating) This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating~ This test shall be performed in accordance with procedure III. Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

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4.7.3 **Performance.** The selected pulse shapers shall be tested as follows: Connect test equipment (items 1 through 4, Table IV) to pulse shaper as shown in Figure 1. Adjust pulse generator (item 2, Table XV) to deliver input signal specified in Table I. Use oscilloscope (item 4, Table IV) to examine each of the outputs specified in Table I and check that each meets the requirements of Table I.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Packaging Data Sheet PS10548509.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pulse shaper is used in Subassembly Test Set MX-8048, part of Radar Test Set AN/TPM-22, and in Radar Receiver-Transmitter RT-860, part of Radar Set AN\VPS-2. This assembly shapes, amplifies and distributes the modulator trigger signal. In the test set, it is used to check the performance of the receiver of Radar Set AN/VPS-2.

6.2 Acquisition requirements. Procurement documents shall specify the foilowing:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of conformance for each lot of shipment or product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

Fire Control System
Radar Set AN/VPS-2
Radar Test Set AN/TPM-22
Receiver-Transmitter RT-860/VPS-2
subassembly Test Set MX-8048/TPM-22
Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A678)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-50673A (AR)	12 December 1990

3. DOCUMENT TITLE

SHAPER, PULSE: 10548806

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME US Army ARDEC Standardization & Specification Office	b. TELEPHONE (Include Area Code) (1) Commercial: (201) 724-6674 (2) AUTOVON: 880-6674	
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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